

Module 0: The Math Toolkit

A comprehensive refresher on the mathematics behind Machine Learning.

We will cover Linear Algebra, Geometry, Calculus, and Statistics.

1. Linear Algebra: The Structure of Data

Machine learning models don't see images or text; they see arrays of numbers. Linear algebra is the study of these arrays.

From Real World to Math World



Images

Pixel Grid (Height $\times$ Width $\times$ Color)

$[[255, 0, 0], [0, 255, 0] \dots]$



Text

Embeddings (Feature Vectors)

"Cat" $\rightarrow [1037] \rightarrow [0.2, -0.5, 0.9]$



Audio

Waveform Samples (Time Series)

$[0.0, 0.5, 0.8, 0.3, -0.1 \dots]$

What does a WHOLE DATASET look like?

If we have **N** samples (e.g., $N=1000$ items), we stack them into a larger array:



Image Dataset (3D Tensor)

Shape: (N, Height, Width, 3)



(1000, 64, 64, 3)

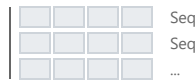
1000 color images, each 64x64 pixels.

$$\mathbf{X} \in \mathbb{R}^{1000 \times 64 \times 64 \times 3}$$

Set: $\mathcal{D} = \{x^{(i)}\}_{i=1}^{1000}$, where $x^{(i)} \in \mathbb{R}^{64 \times 64 \times 3}$

Text Corpus (2D Matrix)

Shape: (N, Sequence_Length)



(1000, 50)

1000 sentences, each up to 50 words long.

$$\mathbf{X} \in \mathbb{R}^{1000 \times 50}$$

Set: $\mathcal{D} = \{x^{(i)}\}_{i=1}^{1000}$, where $x^{(i)} \in \mathbb{R}^{50}$

Audio Batch (2D Matrix)

Shape: (N, Time_Steps)



(1000, 16000)

1000 audio clips, each 1 second long (16k samples).

$$\mathbf{X} \in \mathbb{R}^{1000 \times 16000}$$

Set: $\mathcal{D} = \{x^{(i)}\}_{i=1}^{1000}$, where $x^{(i)} \in \mathbb{R}^{16000}$

Understanding Dimensions

Before we dive into operations, it's crucial to understand the "shape" of our data structures.

Vector Dimension ($\mathbb{R}^n$)

Think of a vector as a single **list of numbers**.

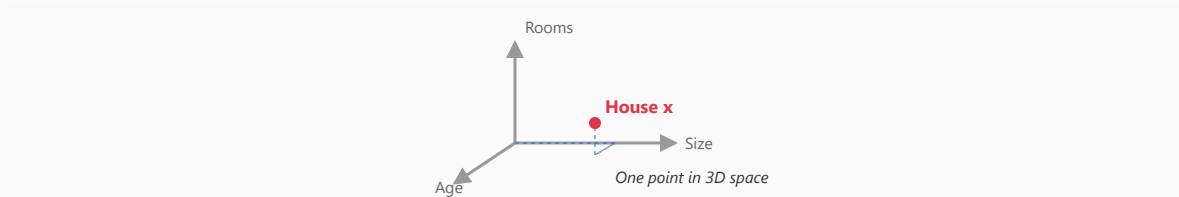
"n-dimensional" just means the list has **n items** in it.

shape = (n,) or (n, 1)

Example: A single house with 3 features (Size, Rooms, Age).

$$\mathbf{x} = \begin{bmatrix} 2000 \\ 3 \\ 15 \end{bmatrix} \in \mathbb{R}^3$$

Visualizing in 3D Space



Matrix Dimension ($\mathbb{R}^{m \times n}$)

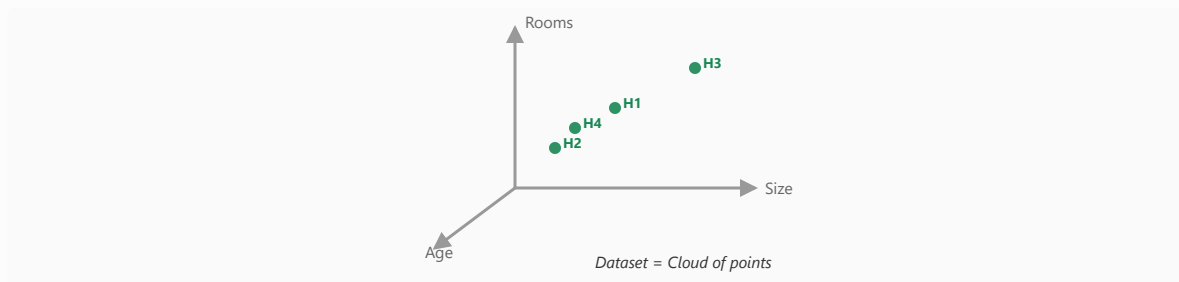
A matrix has **2 dimensions**: m rows (samples) and n columns (features).

shape = (m, n)

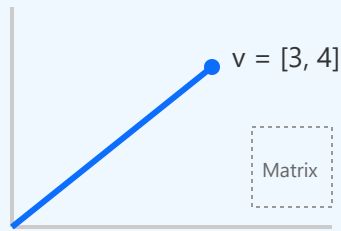
Example: A dataset of 4 houses, each with 3 features.

$$\mathbf{X} = \begin{bmatrix} 2000 & 3 & 15 \\ 1500 & 2 & 10 \\ 2400 & 4 & 5 \\ 1800 & 3 & 20 \end{bmatrix} \in \mathbb{R}^{4 \times 3}$$

Visualizing the Dataset



Vectors & Matrices



Vector ($\mathbf{x}$): A single list of features. E.g., a house.

Matrix ($\mathbf{X}$): A table of data. Rows are samples, columns are features.

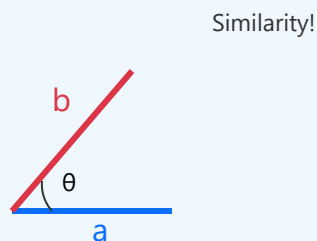
Worked Example

Data: 2 Houses with 2 features (Size, Rooms).

$$\mathbf{X} = \begin{bmatrix} 2000 & 3 \\ 1500 & 2 \end{bmatrix}$$

×

The Dot Product



The fundamental operation of ML. It measures **similarity** or **alignment**.

$$\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i = ||\mathbf{a}|| ||\mathbf{b}|| \cos(\theta)$$

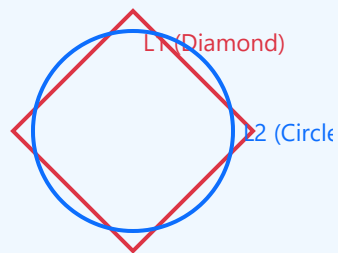
Worked Example

$\mathbf{u} = [1, 5]$ (User), $\mathbf{m} = [1, 5]$ (Movie).

$\mathbf{u} \cdot \mathbf{m} = 26$ (High match!)



Vector Norms (Length)



How "big" is a vector? Critical for **Regularization** (keeping models simple).

- **L2 Norm (Euclidean):** $\|\mathbf{x}\|_2 = \sqrt{x^2 + y^2}$
Represents the straight-line distance. The equation $x^2 + y^2 = r^2$ defines a **Circle**. This smooth shape encourages small weights but rarely zeroes them out.
- **L1 Norm (Manhattan):** $\|\mathbf{x}\|_1 = |x| + |y|$
Represents "taxi" distance (grid movement). The equation $|x| + |y| = r$ defines a **Diamond** (a square rotated 45°). Because this shape has "sharp corners" on the axes, minimizing it often drives coefficients to exactly zero (feature selection).



Eigenvalues & Eigenvectors

Only Stretched!
No Rotation

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

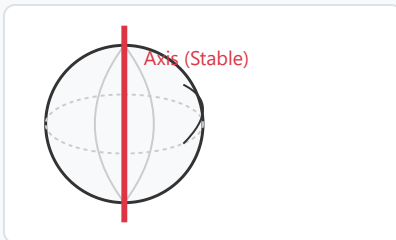
When a matrix transforms a vector, sometimes it just **stretches** it without rotating it.

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Simple Analogy:

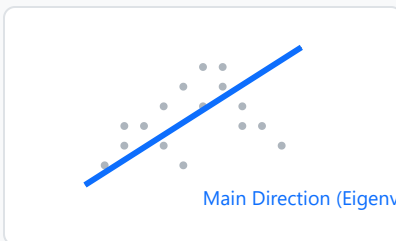
1. The Spinning Globe:

Most points move, but the Axis (Eigenvector) stays still.



2. The Football Shape:

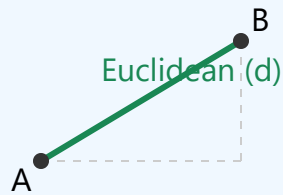
The "Long Axis" of the data cloud is the main Eigenvector.



2. Geometry: Distances & Separations

Algorithms like KNN and SVM rely entirely on geometric concepts.

Distance Metrics



How far apart are two data points? (KNN uses this).

Euclidean Distance:

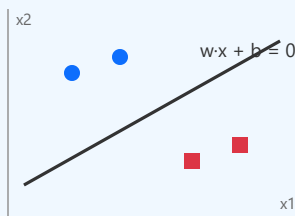
$$d(p, q) = \sqrt{\sum (p_i - q_i)^2}$$

Example (3D): Let $\mathbf{p} = [1, 5, 2]$ and $\mathbf{q} = [3, 2, 8]$.

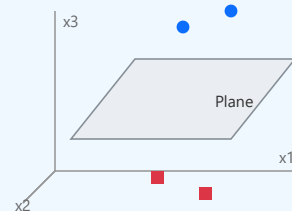
$$d = \sqrt{(1-3)^2 + (5-2)^2 + (2-8)^2}$$

Hyperplanes
 $= \sqrt{(-2)^2 + (3)^2 + (-6)^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$

2D Space (Line)



3D Space (Plane)



A geometric boundary that separates classes.

- **2D Space:** A Line. Equation: $w_1x_1 + w_2x_2 + b = 0$
- **3D Space:** A Flat Sheet (Plane). Equation: $w_1x_1 + w_2x_2 + w_3x_3 + b = 0$
- **General (nD):** A Hyperplane. Equation: $\mathbf{w} \cdot \mathbf{x} + b = 0$

Expanded: $w_1x_1 + w_2x_2 + \dots + w_nx_n + b = 0$

Matrix Form:
$$\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}^T \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + b = 0$$

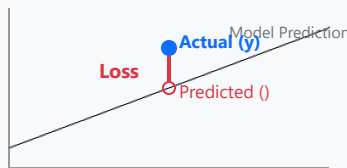
3. Calculus: The Optimization Engine

How do we learn? By minimizing error. Calculus tells us which way to move.

The Golden Rule of Learning (Gradient Descent):

$$w_{new} = w_{old} - \alpha \nabla L(w)$$

(New Weight = Old Weight - Learning Rate $\times$ Gradient)

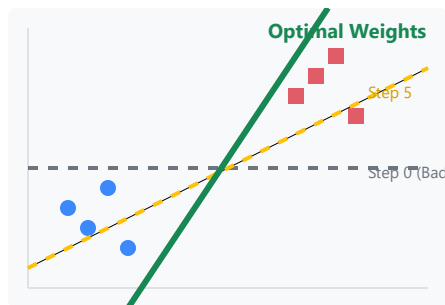


What is "Loss" $L(w)$?

Loss is simply the **distance** (error) between what the model predicts and the actual truth.

- **Large Gap:** High Loss $\rightarrow$ Bad Model (Need to update weights).
- **Small Gap:** Low Loss $\rightarrow$ Good Model.

VISUALIZING WEIGHT UPDATES

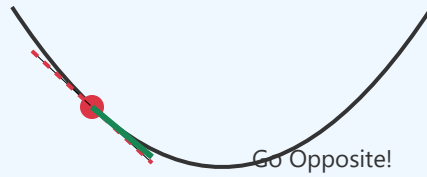


What happens when we update weights?

In classification, weights ($\mathbf{w}$) define the **boundary line**. Updating weights rotates and shifts this line.

- **Step 0 (Grey):** Random weights give a bad line (many errors).
 $w = [0.0, 1.0]$ (Horizontal)
- **Step 5 (Yellow):** Gradient descent pushes the line closer to the gap.
 $w = [0.5, 1.0]$ (Tilting)
- **Optimal (Green):** The weights now perfectly separate the two classes.
 $w = [1.5, 1.0]$ (Perfect Slope)

The Gradient (∇)



The gradient is the steepness. To minimize error, we go **downhill** (opposite the gradient).

Worked Example

Loss Function: $L(w) = w^2$

Derivative: $\frac{dL}{dw} = 2w$

If we are at $w = 3$:

- Gradient is $2(3) = 6$ (Positive slope).
- We need to go **Left** to reduce L .

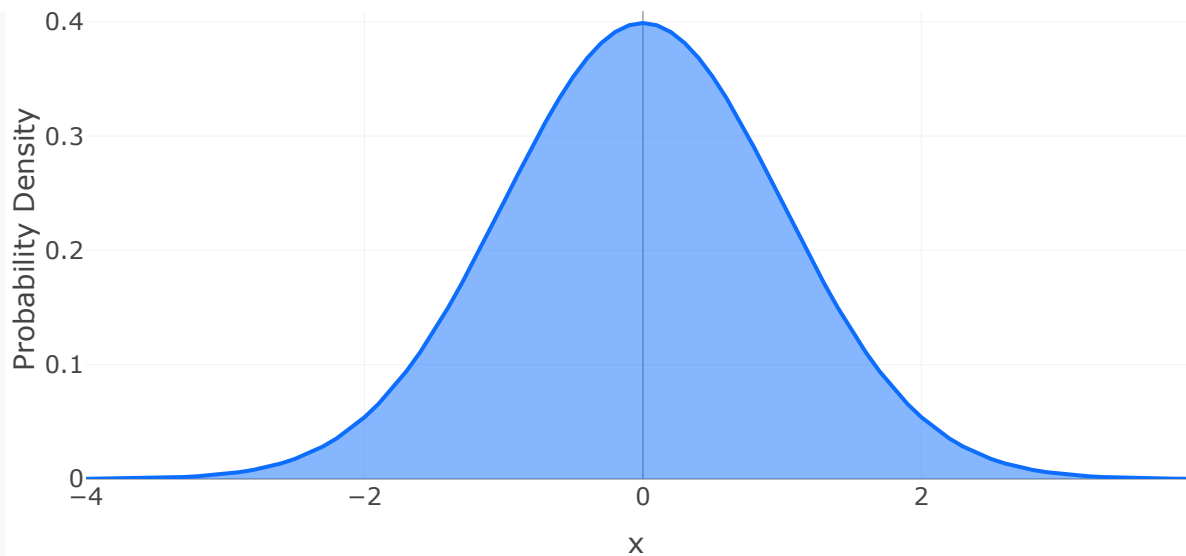
4. Statistics & Probability

Modeling uncertainty and understanding data distributions.

Visualizing Data Distributions

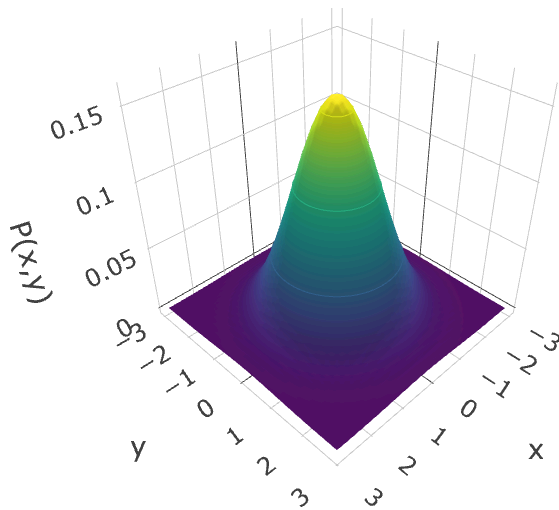
Before calculating stats, it's helpful to visualize how data is spread. The most common pattern is the **Gaussian (Normal) Distribution**.

1 Variable (2D Bell Curve)



Probability Density Function of a single variable x .

2 Variables (3D Surface)



Joint distribution of two variables x and y .

Click and drag to rotate!

Equation (1 Variable)

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ (Mu): Mean (center of the peak)
- σ (Sigma): Standard Deviation (width of the bell)
- x : The variable we are measuring

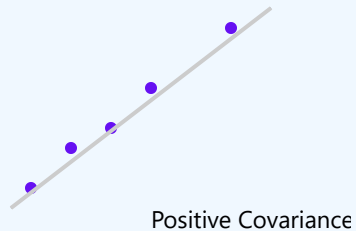
- e : Euler's number (approx 2.718)

Equation (2 Variables, $n = 2$)

$$P(\mathbf{x}) = \frac{1}{2\pi|\Sigma|^{1/2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

- $\mathbf{x} = [x, y]^\top$: Variable vector
- $\boldsymbol{\mu} = [\mu_x, \mu_y]^\top$: Mean vector
- $\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}$: Covariance Matrix
- ρ : Correlation coefficient between X and Y

Mean, Variance, Covariance



- **Mean (Expectation):**

$$\mathbb{E}[X] = \mu = \frac{1}{N} \sum x_i$$

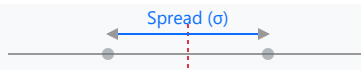
The "Center of Mass" of the data.



- **Variance:**

$$Var(X) = \sigma^2 = \mathbb{E}[(X - \mu)^2] = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Average squared distance from center (Spread).

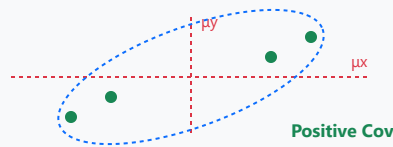


- **Covariance:**

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_x)(Y - \mu_y)]$$

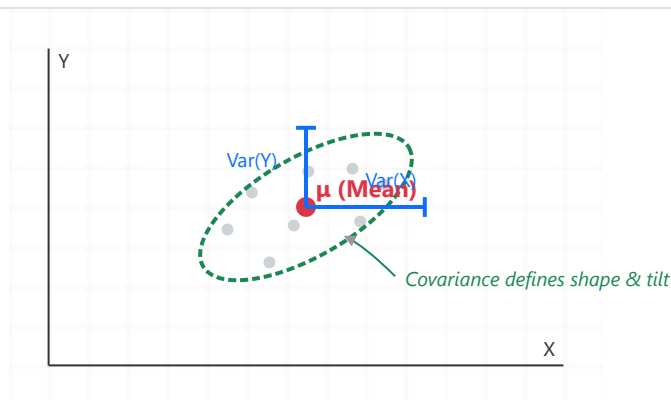
$$= \frac{1}{N} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)$$

How two variables move together (Correlation).

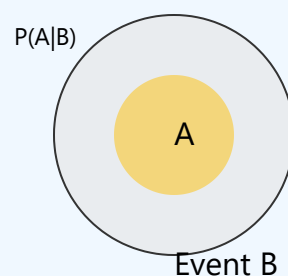


Putting it all together

Visualizing Mean, Variance, and Covariance simultaneously on a 2D dataset.



Probability Basics



Conditional Probability: The chance of A happening within the world where B has already happened.

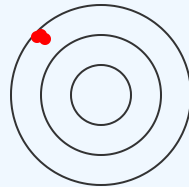
Example: Rain & Clouds

- **P(Rain):** Low (It doesn't rain often).
- **P(Rain | Cloudy):** High (If we *know* it's cloudy, rain is much more likely).

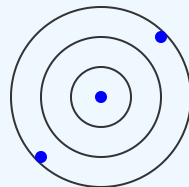
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

5. Statistical Learning Theory

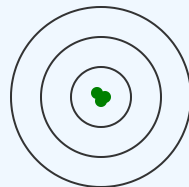
The Bias-Variance Decomposition



High Bias
(Consistent Wrong)



High Variance
(Scattered)



Low Bias/Var
(Bullseye)

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

De-mystifying the Equation

The Left Side (What we measure):

- y : The **True Value** (what we want).
- $\hat{f}(x)$: The **Prediction** (what the model gives).
- $E[\dots]$: The **Expected (Average)** Squared Error.

The Right Side (The Sources of Error):

1. **Bias² (Assumption Error)**: The error from wrong assumptions (e.g., thinking data is linear when it's curved). "*Underfitting*".
2. **Variance (Sensitivity Error)**: The error from being too sensitive to noise in the training data. "*Overfitting*".
3. **σ^2 (Irreducible Error)**: The noise in the real world that no model can fix.

The Trade-off: You can't make both Bias and Variance zero. Making a model more complex lowers Bias but raises Variance. The goal of ML is to find the "Sweet Spot" in the middle!

Now you have the tools!